

Report of Calibration
GLOBAL CALIBRATION LABORATORY at
FLUKE BIOMEDICAL
2 Science Road
Glenwood, IL 60425-1586 USA

CUSTOMER ASSET NUMBER:

MANUFACTURER: FLUKE BIOMEDICAL

MODEL NUMBER: INCU II

SERIAL NUMBER: 45260022

CUSTOMER INFORMATION

NEW UNIT

PROCEDURE USED: INCU-II VER - Exp Unc

TECHNICIAN: Jajuan Jeffries

TEMPERATURE: 23.82 °C

HUMIDITY: 47.4 %rh

SERVICE ORDER NUMBER: 64024

CALIBRATION DATE: 17 JUL 2019

REQUESTED RECALIBRATION DATE: 17 JUL 2020

AS-FOUND STATUS: N/A

AS-LEFT STATUS: Pass

Fluke Biomedical certifies that the above listed instrument meets or exceeds all specifications as stated in the referenced procedure unless otherwise noted. It has been calibrated using measurement standards traceable to the International System of Units (SI) through the National Institute of Standards and Technology (NIST), or other national metrology institutes.

The Fluke Biomedical quality system complies with ISO 17025-2005. Although the item calibrated may meet specification and performance criteria at the time of calibration, due to any number of factors the recommended due date of this item does not imply continuing conformance to specifications during the recommended interval.

This report may not be reproduced, except in full, unless permission for the publication of an approved abstract is obtained in writing from the calibration organization issuing this report.

This report may contain results of tests for which no NIST calibration standard exists, such as oxygen saturation (SpO2). These tests are conducted using in-house performance standards and accepted test procedures.

Any Test Uncertainty Ratio (TUR) that is less than four-to-one will appear under the "TUR" heading on the data record. If the TUR meets or exceeds four-to-one, the field is left blank. Test status is PASS unless noted otherwise.

REMARKS:

APPROVED BY:



Richard Abbott
Technical Services Manager

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STANDARDS USED

ID	Description	Due Date
IRM-1659	TEMPERATURE DRYWELL	Cal Not Required
IRM-1688	CHUB-E4 THERMOMETER	12 Feb 2020
IRM-1691	TEMPERATURE PROBE	13 May 2020
IRM-1773	CALIBRATOR	16 Apr 2020
IRM-1959	SOUND LEVEL METER	03 Dec 2019
IRM-2149	Platinum Resistance Thermometer	30 May 2020
IRM-2154	MICROCLIMATE CHAMBER	Cal Not Required
IRM-2172	ANEMOMETER PROBE	29 Apr 2020
IRM-2231	TEMPERATURE AND HUMIDITY METER	28 May 2020

MODEL NUMBER: INCU II SERIAL NUMBER: 45260022
TEST RESULT: AS LEFT

FLUKE.

Biomedical

Test Description	Expected Value	Measured Value	Lower Limit	Upper Limit	Units	Status	TUR
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UNIT TESTED: INCUII,1.09 ,SN:45260022

Probe Constants Stored Onboard INCU-II

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PROBE NAME CF1 CF2 CF3

T1 Air (RD) = -406606,418829,200023

T2 Air (YW) = -406108,417873,204559

T3 Air (WH) = -407447,419650,197616

T4 Air (BL) = -407572,420030,197425

T5 Air (GN) = -406995,419367,197246

R1 Puck (RD) = -406848,418629,203644

R2 Puck (YW) = -407951,420774,193536

R3 Puck (WH) = -407252,419420,199650

R4 Puck (BL) = -407766,420369,195424

R5 Puck (GN) = -407100,419107,201282

PROBE NAME CF1 CF2 CF3 CF4

Humidity = 3653,3370,1104972416,922529408

PROBE NAME CF1 CF2

Sound = 2559,3739

PROBE NAME CF1 CF2 CF3 CF4 CF5 CF6

Airflow = 60614,54731,50148,47399,45632,44278

CF7 CF8 CF9 CF10 CF11

43240,42289,41409,40645,39877

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AIR CONVECTION TEMPERATURE PROBE ACCURACY TESTS

25 degC Setpoint

Temp Probe T1	24.994	24.99	24.94	25.04	°C	1.0
Temp Probe T2	24.994	24.99	24.94	25.04	°C	1.0
Temp Probe T3	24.993	24.98	24.94	25.04	°C	1.0
Temp Probe T4	24.994	25.00	24.94	25.04	°C	1.0
Temp Probe T5	24.994	24.99	24.94	25.04	°C	1.0

2 degC Setpoint

Temp Probe T1	1.976	1.97	1.93	2.03	°C	1.0
Temp Probe T2	1.974	1.97	1.92	2.02	°C	1.0
Temp Probe T3	1.974	1.96	1.92	2.02	°C	1.0
Temp Probe T4	1.975	1.98	1.93	2.02	°C	1.0
Temp Probe T5	1.975	1.97	1.93	2.02	°C	1.0

48 degC Setpoint

Temp Probe T1	47.987	47.99	47.94	48.04	°C	1.0
Temp Probe T2	47.986	47.99	47.94	48.04	°C	1.0
Temp Probe T3	47.986	47.98	47.94	48.04	°C	1.0
Temp Probe T4	47.988	47.98	47.94	48.04	°C	1.0
Temp Probe T5	47.988	47.98	47.94	48.04	°C	1.0

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2 Science Road Glenwood, IL 60425-1586 USA

Telephone

833.296.9240

Internet

www.flukebiomedical.com

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MODEL NUMBER: INCU II SERIAL NUMBER: 45260022

TEST RESULT: AS LEFT

Test Description	Expected Value	Measured Value	Lower Limit	Upper Limit	Units	Status	TUR
AIR CONVECTION TEMPERATURE PUCK ACCURACY TESTS							
39 degC Setpoint							
39 degC Setpoint							
Temp Puck R1	39.231	39.25	39.03	39.43	°C		2.9
Temp Puck R2	39.231	39.26	39.03	39.43	°C		2.9
Temp Puck R3	39.231	39.23	39.03	39.43	°C		2.9
Temp Puck R4	39.232	39.22	39.03	39.43	°C		2.9
Temp Puck R5	39.224	39.21	39.02	39.42	°C		2.9
HUMIDITY PROBE ACCURACY TESTS							
HUMIDITY PROBE ACCURACY TESTS							
60.0 %RH	63.78	61.8	60.8	66.8	%RH		1.4
30.0 %RH	34.00	33.5	31.0	37.0	%RH		1.7
SURFACE TEMPERATURE ACCURACY TESTS							
mV-Simulated Type-K Thermocouple							
2.00 °C	2.000	2.10	1.50	2.50	°C		2.6
25.00 °C	25.000	25.10	24.50	25.50	°C		2.6
58.00 °C	58.000	58.10	57.50	58.50	°C		2.6
AIRFLOW ACCURACY TESTS							
0.30 m/s	0.300	0.40	0.20	0.40	m/s		3.5
0.50 m/s	0.510	0.58	0.41	0.61	m/s		3.5
SOUND PRESSURE LEVEL ACCURACY TESTS							
60 dB(A)	60.0	58	55	65	dB(A)		2.1
80 dB(A)	80.0	77	75	85	dB(A)		2.1
K-TYPE THERMOCOUPLE PROBE FUNCTIONALITY TEST							
K-Type Thermocouple Probe is functional							

*** END_OF_REPORT ***

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